

Problem

A factory has the following four major loads:

- A motor rated at 5 hp, 0.8 pf lagging (1 hp = 0.7457 kW).
- A heater rated at 1.2 kW, 1.0 pf.
- Ten 120-W lightbulbs.
- A synchronous motor rated at 1.6 kVAR, 0.6 pf leading.

(a) Calculate the total real and reactive power.

(b) Find the overall power factor.

Step-by-step solution

Step 1 of 7

(a)

$$P_1 = 5 \times (0.7457 \times 10^3)$$

$$P_1 = 3.7285 \text{ k W}$$

$$S_1 = \frac{P_1}{\text{pf}}$$

$$S_1 = \frac{3.7285 \times 10^3}{0.8}$$

$$S_1 = 4.661 \text{ kVA}$$

Step 2 of 7

$$Q_1 = S_1 \sin(\cos^{-1}(0.8))$$

$$Q_1 = 2.796 \text{ k VAR}$$

$$\therefore S_1 = 3.7285 + j2.796 \text{ k V A}$$

Step 3 of 7

$$P_2 = 1.2 \text{ k W} \quad \&$$

$$Q_2 = 0 \text{ VAR}$$

$$\therefore S_2 = 1.2 + j0 \text{ k V A}$$

Step 4 of 7

$$P_3 = (10)(120)$$

$$P_3 = 1.2 \text{ k W}$$

$$Q_3 = 0 \text{ VAR}$$

$$\therefore S_3 = 1.2 + j0 \text{ k VA}$$

Step 5 of 7

$$Q_4 = 1.6 \text{ k VAR}$$

$$\cos \theta_4 = 0.6$$

$$\sin \theta_4 = 0.8$$

$$S_4 = \frac{Q_4}{\sin \theta_4}$$

$$S_4 = 2 \text{ k VA}$$

$$P_4 = S_4 \cos \theta_4$$

$$P_4 = (2 \times 10^3) \times (0.6)$$

$$P_4 = 1.2 \text{ k W}$$

$$\therefore S_4 = 1.2 - j1.6 \text{ k VA}$$

Step 6 of 7

Therefore,

$$S = S_1 + S_2 + S_3 + S_4$$

$$S = 7.3285 + j1.196 \text{ k VA}$$

Total real power = 7.3285 k W

Total reactive power = 1.196 k VAR

Step 7 of 7

(b)

$$\theta = \tan^{-1} \left(\frac{1.196}{7.3285} \right)$$

$$\theta = 9.27^\circ$$

Therefore,

$$\text{P f} = \cos \theta$$

$$\text{P f} = 0.987$$